

Analysis and Design of Algorithms

Midterm 2 · Problems

Problem 1: Electronic Circuit Optimization (3.5 points)

A company specializing in the development of electronic circuits has hired you to optimize the design of its circuits. Your first task consists of improving the design of the most basic circuits, which consist of N pins connected to each other by cables. Each cable exclusively connects a pair of distinct pins, and the company guarantees that its current designs, although not optimized, always allow every pin to be reached from any other pin, forming a fully connected circuit.

The objective is to minimize the total amount of cable needed to maintain this connectivity between the pins. It is important that the optimized designs preserve the full-connectivity property.

You are asked to do the following:

a) (2 points)

Describe the data structures needed to model this problem. Define the relevant variables and provide a graphical representation of these structures.

b) (1 point)

State known algorithms that solve this optimization problem. Briefly describe each algorithm and justify its suitability for meeting the company's requirements.

d) (3 points)

Implement an algorithm that, given the optimized circuit resulting from the previous section, identifies the two pins whose distance, defined as the length of the shortest path between them, is maximum. The algorithm must return this distance and the intermediate pins that make up the shortest path between the corresponding pair of pins. The code must include comments explaining how it works. Show an example call to the algorithm from a main program, specifying the initial parameter values.

Note: section c) of the original exam was not used in the consistency study due to PDF scanning issues.